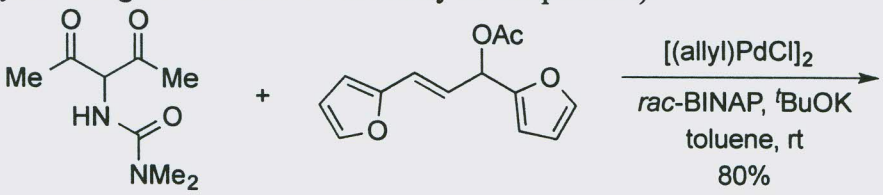
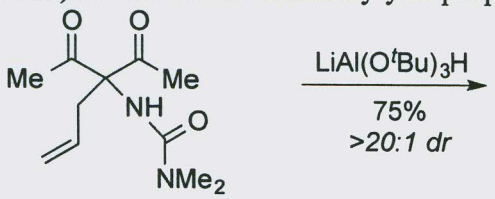
Organic Cume
04-25-2014

Professor Jeffery S. Johnson, a recent Departmental speaker, discussed the synthesis of Pactamycin in his talk (JACS **2013**, *135*, 17990-17998). The following four reactions are taken out of their synthetic studies of Pactamycin.

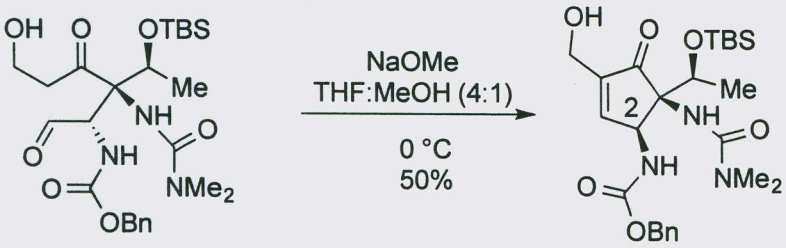
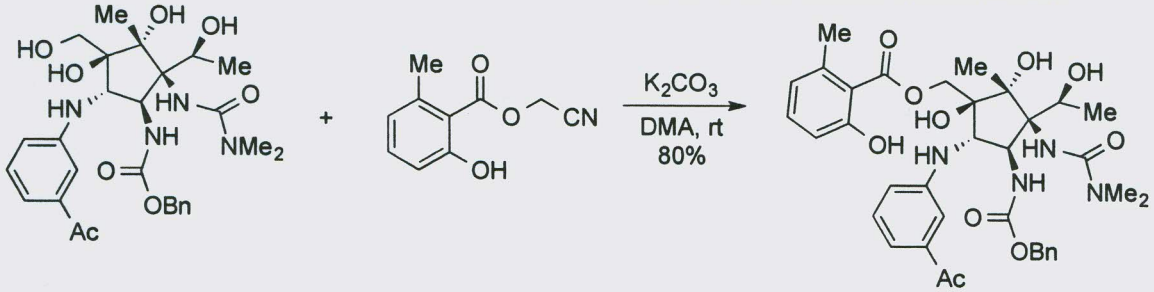
1. Give the product and write a reasonable mechanism to account for its formation (Note: you can ignore the stereochemistry of the product).



2. Give the product and specify its stereochemistry. Provide a rational (e.g., transition state) for the stereochemistry you propose.



3. Write a reasonable mechanism for the following transformation. Your mechanism should account for the epimerization at C2.



4. Give the product and write a reasonable mechanism to account for its formation.